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Effectiveness of photobiomodulation therapy on pain intensity in postpartum women with nipple or perineal trauma: protocol for a multicentre, double-blinded, parallel-group, randomised controlled trial

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ABSTRACT

Introduction Photobiomodulation (PBM) using low-level laser can affect tissue repair mechanisms and seems promising in reducing pain intensity. However, few studies support the effectiveness of PBM on postpartum period complications, such as nipple and/or perineal trauma and pain, probably due to the low doses used. The primary objective of this study is to analyse the effectiveness of PBM on pain intensity in the nipple and perineal trauma in women in the immediate postpartum period. Secondary objectives are to evaluate the effect on tissue healing and the women's satisfaction.

Methods and analysis A double-blind, multicentre, parallel-group, randomised controlled trial will be performed in two public referral maternity hospitals in Brazil with 120 participants, divided into two arms: 60 participants in the nipple trauma arm and 60 participants in the perineal trauma arm. Participants will be women in the immediate postpartum period, who present with nipple trauma or perineal trauma and report pain intensity greater than or equal to 4 points on the Numerical Rating Scale for Pain. Block randomisation will be performed, followed by blinding allocation. In the experimental group, one application of PBM will be performed between 6 hours and 36 hours after birth. For the sham group, the simulation will be carried out without triggering energy. Both participants and the research evaluator will be blinded to the allocation group. Intention-to-treat method and the between-group and within-group outcome measures analysis will be performed.

Ethics and dissemination This research protocol was approved by the Research Ethics Committees of the University of Campinas, Brazil, and of the School Maternity Assis Chateaubriand, Brazil (numbers CAAE: 59400922.1.1001.5404; 59400922.1.3001.5050). Participants will be required to sign the informed consent form to participate. Results will be disseminated to the health science community.

Trial registration number Brazilian Registry of Clinical Trials (RBR-2qm8jrp).

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ The novelty of this trial is the power used of 100 mV (other studies used devices with lower powers) and new energy parameters (2 J of red light and 4 J of infrared light per point).
⇒ Many women experience nipple or perineal pain very early on, which compromises breast feeding and their recovery in general, and this may be a non-pharmacological option for relieving this pain.
⇒ As a limitation, we cite the impossibility of applying the laser to the studied population after hospital discharge due to sociodemographic issues.
⇒ Another limitation will be the late assessment after the intervention (after 7–10 days) being carried out via telephone interview, which prevents an objective assessment of healing.
⇒ The use of laser for both nipple and perineal injuries is easy, low cost, safe and well accepted by women in the postpartum period.

INTRODUCTION

Photobiomodulation (PBM), also known as low-level light therapy, uses low-power light sources such as laser (Light Amplification Stimulated Emission Radiation) or Light Emitting Diodes in therapeutic applications.[1] It is widely used in several areas of health, including dermatology, dentistry, rheumatology and physiotherapy.[2 3]

PBM in the red (600–700 nm) and near-infrared (770–1200 nm) spectrum is the most studied, as its drug-free, non-invasive, safe actions promote tissue repair, reduce the inflammatory process and offer analgesia, based on non-ionising radiation and induction of non-thermal responses. Biological responses to PBM stimulation depend on a

large variety of parameters, of which the dose of energy used is one of the most important.[1–3]

The immediate postpartum period is considered critical for women's and newborns' health, as this is when major complications might occur. Nipple (NT) and perineal trauma (PT) can interfere during this period if pain intensity leads to limitations on women's daily activities, breast feeding and newborn care.[4]

PT occurring during vaginal birth, whether spontaneous (perineal tears) or surgical (episiotomy), can generate postpartum perineal pain and contribute to pelvic floor dysfunctions, such as sexual dysfunction and urinary and/or anal incontinence.[5] For perineal pain relief in the immediate postpartum period, analgesics, cryotherapy and transcutaneous electrical neurostimulation have been used.[6–8] Recent studies have investigated the effects of PBM on pain relief and healing after episiotomy in postpartum women, using total doses of 5 and 8.8 J/cm^2. Although the results did not prove significant benefit, studies suggest the use of different dose parameters in future research.[9 10]

NT is defined as a painful lesion in the areola and nipple integument, which can be a fissure, erosion or ulceration of the skin, with clinical signs of erythema, oedema, blisters, ecchymoses and bleeding.[11] Pain restricts milk production and secretion, causes maternal stress, inhibits oxytocin production and hinders breast feeding. Lanolin is the most widely recommended intervention. PBM appears promising in reducing pain intensity and promoting healing and warrants further investigation.[12]

Although PBM has been incorporated into some clinical practices for the treatment of postpartum complications, the literature on the subject is limited. Well-designed research is needed to evaluate its effects on pain relief. This study aims to fill this gap in the scientific literature and contribute to advances in postpartum care to improve quality of life. For this, it was decided to compare the experimental group with a sham group (simulation).

PBM is an easy-to-apply therapy that has shown promising results in other clinical situations, which justifies interest in its use in the NT and PT treatment. Investing in this therapeutic option can reduce unfavourable situations such as puerperal pain and early weaning, thus improving mothers' quality of life.

OBJECTIVES

The primary objective is to evaluate the effectiveness of PBM on pain intensity in women with PT or NT in the immediate postpartum period. The secondary objectives are to evaluate the women's perception of perineal or nipple healing and their satisfaction with the intervention.

TRIAL DESIGN

This is a randomised controlled trial, multicentre, double-blinded, parallel group, with one experimental group and one sham group in a 1:1 allocation. We chose to carry out the control with a sham group (simulated application of PBM therapy) to minimise courtesy bias and the Hawthorne effect.[13] This protocol is reported following the Standard Protocol Items: Recommendations for Interventional Trials checklist.[14]

METHODS

Study setting

The study setting will be two centres in different regions of Brazil: State University of Campinas Women's Hospital, in Southeast Brazil, and School Maternity Assis Chateaubriand, in Northeast Brazil. Both are public referral centres in maternal–infant care; together they assist more than 6000 births per year in habitual risk and high-risk pregnant women.

Eligibility criteria

PT arm

For the PT arm, a prescreening will be carried out using medical records to check the postpartum time in hours and the occurrence and classification of PT. Women meeting the criteria will be approached by the researchers. Inclusion criteria for the PT arm will be age 18 years or above, PT resulting from childbirth (episiotomy or laceration of at least second degree), reported perineal pain of at least 4 on the Numerical Rating Scale for Pain (NRS Pain) at the time of the approach, understanding of Portuguese and cognitive capacity for consent. Exclusion criteria for the PT arm will be active gynaecological infections, unresolved puerperal haemorrhage, or hearing or cognitive impairment.

NT arm

For the NT arm, women within the postpartum period 6–36 hours (data from medical records) will be approached by the researchers and asked about the presence of NT, which will be confirmed through researcher inspection. Inclusion criteria for the NT arm will be women aged 18 years or above who are breast feeding and have NT, report nipple pain of at least 4 on the NRS Pain at the time of the approach, understand Portuguese and have the cognitive capacity for consent. Exclusion criteria for the NT arm will be mastitis or neoplasia, photosensitivity to sunlight, breast malformation, breast implant or breast reduction surgery, or hearing or cognitive impairment.

Interventions

In each arm (PT and NT), eligible participants, after assessment by a blinded investigator, will be randomised between the experimental and sham group. Participants will be approached at 6–36 hours after childbirth for baseline assessment and application of only one intervention.

For the application of PBM, the Therapy EC device (DCM brand) will be used as an irradiation source (registration at the National Health Surveillance Agency–ANVISA no. 80030810156; certified by the National Institute of Metrology, Quality, and Technology) (online

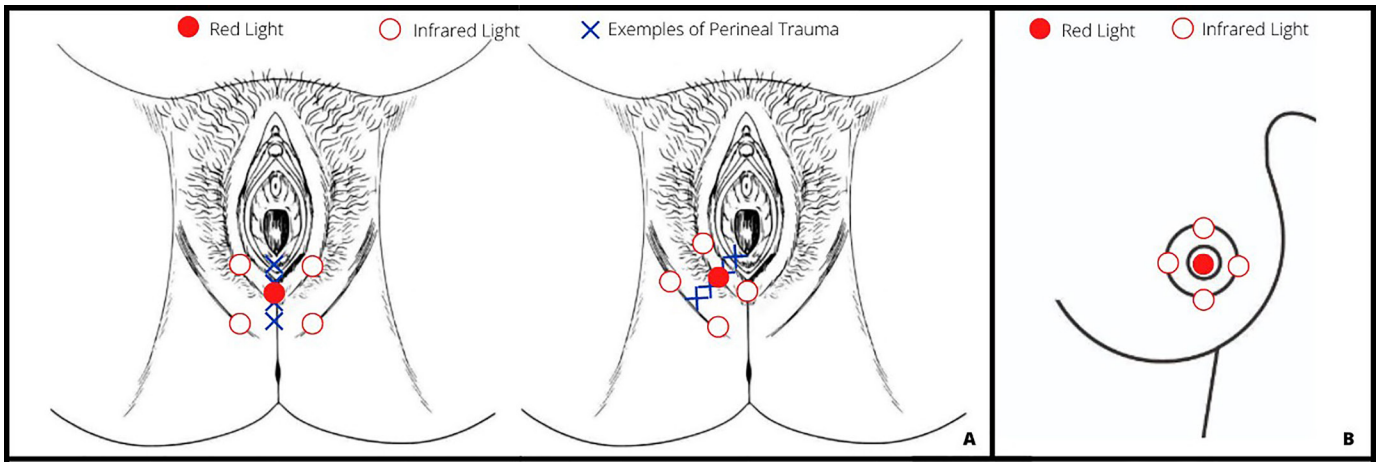


Figure 1 (A) Perineal trauma arm application points demonstration; (B) nipple trauma arm application points demonstration.

supplemental figure). The laser device emits 1 J every 10 s. A condom will protect the device for application and will be changed for each patient. Condom use causes a 20% loss of potency (assessed and measured by the device manufacturer), but it is necessary for patient security.[15]

All participants in the PT arm will be in a gynaecological position on their beds, with a sheet covering their legs, and will wear protective glasses. The application assistant will wear protective glasses, position the laser device perpendicular (90°) to the point that will receive an application, touch the pointer to the skin, cover the region with her hand and apply (or simulate) the PBM.[9 10] There will be one or more application points on the PT site with 2 J of red light (wavelength 660±10 nm) and four or more application points around the site with 4 J of infrared light (wavelength 808±10 nm), with 2 cm distance between application points (number of points will depend on the trauma's extension) (figure 1A). Application time will total at least approximately 180 s.

In the NT arm, all participants will be positioned seated on their beds, with their feet supported and spine erect, wearing protective glasses and facing away from the breast receiving the intervention. The application assistant, wearing goggles, will position the laser device perpendicular (90°) to the point that will receive an application, touch the pointer to the skin, cover the region with her hand and then apply (or simulate) the PBM.[16] There will be one application point on the nipple trauma site with 2 J of red light (wavelength 660±10 nm) and four application points around the nipple, at cardinal points with a 2 cm distance between application points (figure 1B), with 4 J of infrared light (wavelength 808±10 nm), totalling 180 s of application.

The depth of absorption of red light is more superficial, having been more used for wound healing. Infrared light, on the other hand, is absorbed more deeply and may affect nociceptors, causing pain relief.[17 18] The new parameters of potency and amount of energy (2 J of red light and 4 J of infrared light) were based on previous studies,[16 19] and the association of these two ways of application in the trauma (perineal and nipple) is the propose of this study.

In both arms, sham group participants will receive a PBM simulation with the same points, without triggering energy. Participants will be informed that the device will emit a 'beep' regardless of the group. Participants may feel slight heating when the red light is applied to the lesion. The parameter details that will be used are shown in table 1.

At the participant's request, the intervention will be interrupted, and the supported energy (and time in seconds) will be registered; the participant will not be discontinued from the research. Loss to follow-up will occur if participants have a serious medical condition with transfer to the intensive care unit or an adverse reaction to PBM. In any of the above cases, the participant

Table 1 Photobiomodulation therapy parameters

Wavelength (infrared light)	808±10 nm	
Wavelength (red light)	660±10 nm	
Power	100 mW±20%	
Supply voltage	100–240 V~	
Input current	Alternating	
Output current	Continuous	
Input power	25 VA	
Supply frequency	50/60 Hz	
Irradiation area/point	0.04 cm^2	
Session time (after delivery)	6–36 hours	
Session time (after 1st session)	24–48 hours	
Energy	**PT arm**	**NT arm**
Energy/point (red light)	2 J	2 J
Energy/point (infrared light)	4 J	4 J
Energy/session	14 J, at minimum	14 J

NT, nipple trauma; PT, perineal trauma.

will only be discontinued from the research; however, the data collected so far may be included in the analysis. The participant will be removed from the research upon consent withdrawal.

Each participant will be informed about the possible benefits of PBM therapy in pain relief and the possibility of sham group allocation. We will stress to participants the importance of completing the pain diary and responding to the telephone interview.

Participants in both maternity hospitals receive routine postpartum care which includes offers of pharmacological therapies for pain control on participants' demand. We will control this variable by registering the type, number, dose and time of analgesics.

Outcomes

Primary outcome measures

Pain intensity: PT arm

Pain intensity will be assessed by the Short-Form McGill Pain Questionnaire (SF-MPQ) Scale and NRS Pain before, at 30 min after and more than 12 hours after intervention by a blinded investigator. Developed by Melzack in 1987, the SF-MPQ assesses multiple aspects of pain, including 11 sensory and 4 affective descriptors, which are rated on an intensity scale from 0 (none) to 3 (severe), plus the Present Pain Intensity and the Visual Analogue Scale.[20] The Brazilian-Portuguese version of SF-MPQ has demonstrated high levels of internal consistency (Cronbach's alpha range=0.70–0.79), reliability (intraclass correlation coefficient range=0.69–0.85) and agreement (SE of the measurement range=0.80–6.92).[21 22] The NRS Pain[20] when performing daily activities of mobility (sitting, standing, walking, lying down), self-care (urinating, evacuating, bathing, eating, sleeping) and care with the newborn (breast feeding and exchange of newborns) was developed by the authors based on the work of East *et al*.[23] The NRS Pain is a single 11-point numerical scale from 0 (no pain) to 10 (worst pain imaginable).[20] Each participant will record her pain score (using the NRS Pain) in a pain diary form at 1 hour, 3 hours, 6 hours and 12 hours after the application of PBM; pain intensity will also be recorded through a telephone interview at 7–10 days after the intervention, using the NRS Pain.

Pain intensity: NT arm

A blinded investigator will assess pain intensity by the NRS Pain[20] before, 30 min after and more than 12 hours after PBM application. The participant will record her pain score in the specifically prepared pain diary during breast feeding within 12 hours after the application of PBM. A telephone interview 7–10 days after the intervention will elicit information about pain intensity.

Secondary outcome measures

Healing: PT arm

Perineal healing will be assessed using the REEDA Scale, which includes five items related to the healing process that form the acronym REEDA: Redness, Oedema, Ecchymosis, Discharge and Approximation. Each item is evaluated on a scale from 0 to 3, according to the severity of inflammation. The highest score of 15 indicates the worst perineal healing. Various studies have shown the REEDA Scale to be a reliable instrument to assess the inflammatory process and tissue repair of all types of postpartum perineal injury.[24 25]

PT healing will be evaluated using the REEDA Scale before the intervention, 30 min after the intervention and more than 12 hours after the intervention, by the same evaluator. Women who are discharged from the hospital early may not have the evaluation after 12 hours.

In the telephone interview, the participant will be asked about wound healing using a Likert scale (improved a lot OR improved a little OR is the same OR worsened a little OR got a lot worse).

Healing: NT arm

Nipple and areola complex lesions (NALs) will be evaluated using the Instrument for Classification of Nipple-Areolar Lesions (ILMA) (table 2), which will be on the data collection form. The ILMA includes two domains: NALs without interruption of the skin barrier (four items: erythema, ecchymosis, oedema and vesicle) and NALs with interruption (three items: fissure, erosion and crust). This is the first validated instrument to assess NALs resulting from breast feeding.[11]

NT healing will be evaluated using the ILMA Scale before the intervention, 30 min after the intervention and more than 12 hours after the intervention, by the same evaluator. Women who are discharged from the hospital early may not have the evaluation after 12 hours.

In the telephone interview, the participant will be asked about wound healing using a Likert scale (improved a lot OR improved a little OR is the same OR worsened a little OR got a lot worse).

Satisfaction with the intervention

Using a 5-point Likert scale (very satisfied, satisfied, neither satisfied nor dissatisfied, dissatisfied, very dissatisfied), participant satisfaction will be assessed through a telephone interview at 7–10 days after the intervention, for both PT and NT arms. Table 3 shows the measuring instruments that will be used in each step of the flow.

Participant timeline

Figure 2 shows the Consolidated Standards of Reporting Trials diagram, which illustrates the flow of participants throughout the research process, for both the PT arm and NT arm.

Sample size

To calculate the sample size, the mean comparison method between two groups was used in an analytical study with a quantitative variable, setting the alpha significance level or type I error at 5% (alpha=0.05) (or a CI of 95%), the power of the sample at 80% (or type II error at 20% (beta=0.20), and using the mean and SD values in each group, obtained from the literature.

Table 2 Nipple and areola complex lesions classification instrument

Definition of nipple and areola complex lesions	Changes in the characteristics of the nipple and areola complex skin during breast feeding, identified through changes in colour, thickness, fluid content or tissue loss.	
Nipple and areola complex lesions without skin barrier disruption	Erythema	A lesion characterised by a change in the skin's natural colour to red or pink in a particular nipple–areola area.
	Ecchymosis	A lesion characterised by a change in the skin's natural colour to purple or brown in a particular nipple–areola area.
	Oedema	Modification of the skin's natural thickness due to accumulation of fluid (plasma) in the dermis or hypodermis, with the presence of distension, shine and/or skin stiffness.
	Vesicle	A lesion with modification of the skin's natural liquid content, identified by the presence of a bulge filled with clear (serous) or white (milky) liquid, measuring up to 0.5 cm in diameter, located on the nipple surface.
Nipple and areola complex lesions with disruption of the skin barrier	Fissure	Linear and narrow continuity solution, which may present with bleeding or serous fluid of variable depth.
	Erosion	Continuity solution of circumscribed or irregular shape, with loss of epidermis or dermis, which may present with bleeding or serous fluid. When it occurs more superficially, it is called excoriation.
	Crust	Result of the drying of serous, purulent or haematic exudate from a previously injured area. It can be light yellow, brown or dark red.

The sample size calculation was made based on the work by Alvarenga *et al*[9] which compared the mean of perineal pain (measured by NRS), between the experimental (laser group) and sham groups. The mean delta of 0.4 in the experimental group (4.5 (SD 3.0) before, 4.1 (SD 2.8) after) was compared with the mean delta of 0.0 in the control group (2.0 (SD 2.2) before, 2.0 (SD 2.2) after), with an estimated SD of the delta 0.5. This equates to an effect size of around 0.8 (a medium effect size). Considering the significance level was set at 5% and the power of the sample at 80%, it was estimated that a minimum sample of n=52 women with perineal laceration would be representative for comparing the mean pain delta between the two groups after the first intervention. Considering the possibility of losing participants, we seek 60 participants (n=30 in the experimental group and n=30 in the sham group).

To calculate the sample size to compare the mean delta nipple pain (before vs after therapy)[26] and mean nipple pain[16] between the experimental (M=−1.3; SD=0.66) and sham (M=−0.5; SD=0.51) groups, the level of significance was set at 5% and the power of the sample at 80%. This equates to an effect size of around 1.96 (a large effect size). Based on the results, it was estimated that a minimum sample of n=20 women with nipple trauma (n=10 in the control group and n=10 in the laser group) would enable comparing the mean delta of pain between the two groups. However, data from Camargo *et al*[16] indicate this sample size to be unfeasible, due to lack of difference between the control and laser groups. Thus, we seek

Table 3 Measurement instrument at the study timeline

Arms	Baseline assessment	Intervention	Reassessment 30 min after intervention	During the next 12 hours	Reassessment 12 hours after intervention	Telephone interview 7–10 days after birth
PT arm	Sociodemographic data Obstetric data Pain (SF-MPQ/NRS Pain) Healing (REEDA)	PBM therapy or simulation	Pain (SF-MPQ/NRS Pain) Healing (REEDA)	Pain (pain diary with NRS Pain)	Pain (SF-MPQ/NRS Pain) Healing (REEDA)	Pain (NRS Pain) Healing (Likert scale) Satisfaction (Likert scale)
NT arm	Sociodemographic data Obstetric data Pain (NRS Pain) Healing (ILMA)	PBM therapy or simulation	Pain (NRS Pain) Healing (ILMA)	Pain (pain diary with NRS Pain)	Pain (NRS Pain) Healing (ILMA)	Pain (NRS Pain) Healing (Likert scale) Satisfaction (Likert scale)

ILMA, Instrument for Classification of Nipple-Areolar Lesions; NRS Pain, Numerical Rating Scale for Pain; NT, nipple trauma; PT, perineal trauma; REEDA, Redness, Oedema, Ecchymosis, Discharge and Approximation; SF-MPQ, Short-Form McGill Pain Questionnaire.

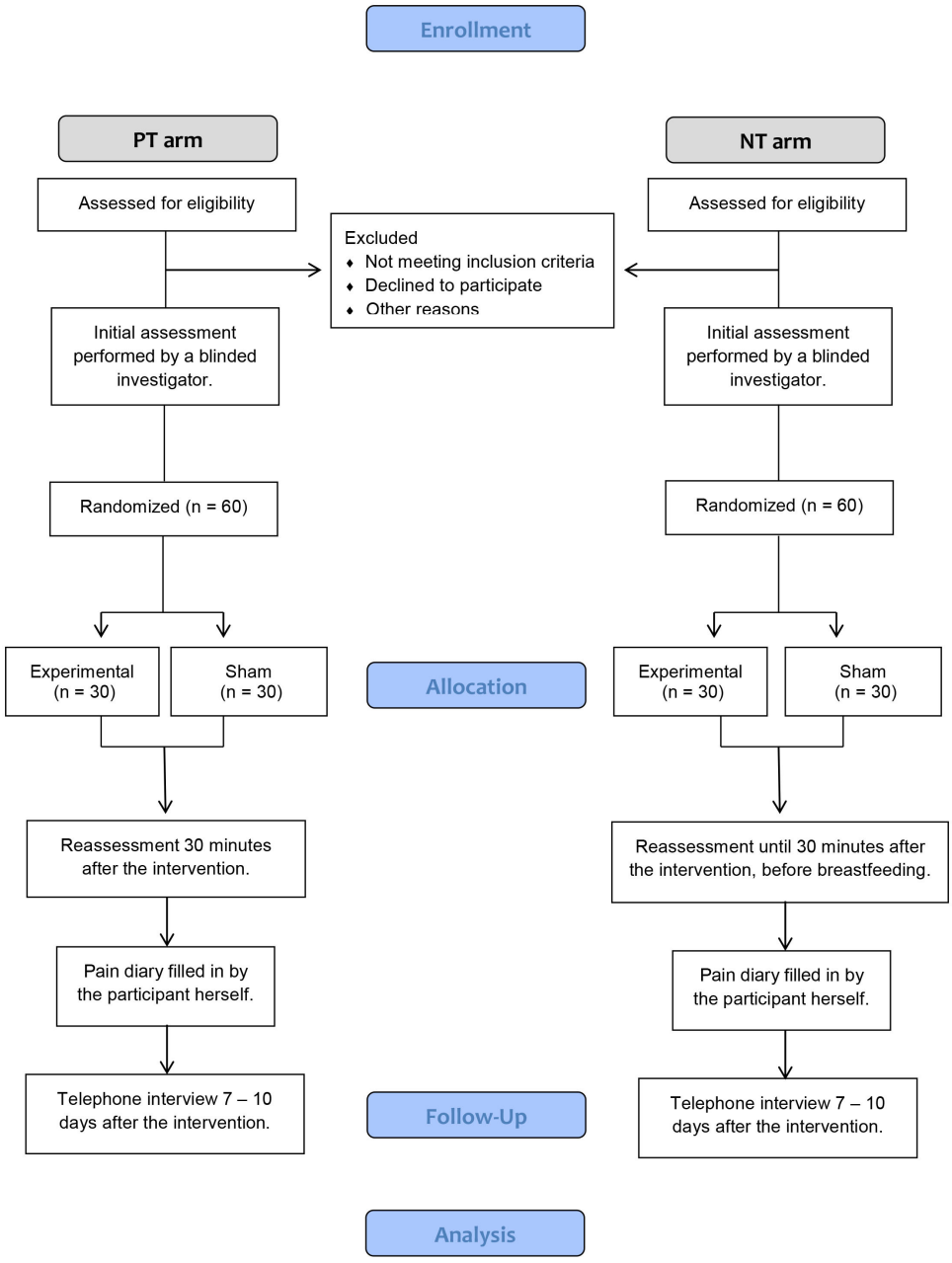


Figure 2 CONSORT diagram of the flow of participants throughout the research process. CONSORT, Consolidated Standards of Reporting Trials; NT, nipple trauma; PT, perineal trauma.

60 participants (30 in the experimental group and 30 in the sham group).

Recruitment

Each centre involved in the PBM puerperium trial was chosen based on the profile of patients it serves and the number of childbirths it records. In addition, the centres are located in different regions of the country, which contributes to a better representation of the Brazilian reality.

Recruitment will be performed by the research member responsible for recruitment, a blind investigator, by checking medical records. Postpartum women will be selected in the rooming-in or postpartum wards of one of the two centres according to eligibility criteria for each arm and invited to participate.

Allocation

Random sequences will be computer generated at http://www.random.org and will be followed for allocation of participants to the experimental or sham group in each arm (NT arm and PT arm), with a 1:1 allocation. The randomisation will be in blocks of 30 participants to ensure a random sequence by each centre. Randomisation will be requested by the research member responsible for recruitment. To ensure concealment of the allocation, sealed and opaque envelopes will be used. To control allocation bias, the envelopes will be opened only

by the application assistant (unblinded), who will not be involved in this study.

Blinding

The study will be double blind, since: (1) the participants will not be aware if they will be in the experimental or sham group; and (2) the assessing researcher will be blinded to the allocation group. Only the application assistant (unblinded), not involved in this study, will know a participant's allocation. Assessments will be conducted by a blinded investigator. The assessor will receive assessment training. A member of the research team will feed data into the computer in separate datasheets so that the researchers can analyse data without having access to information about the allocation.

The purpose of this trial is a single intervention under supervision; if there is any discomfort or adverse effect beyond what was expected, the intervention will be interrupted. If any complications occur after the intervention, the principal investigator (who is not involved in the assessment or intervention) will be contacted.

Data collection, management and analysis

Data collection will begin in February 2023 and will end in December 2023. The data will be collected and stored according to our institutions' data security standards using the REDCap. REDCap allows a direct export as a CSV file, which will then be uploaded to the SPSS program (V.20.0) for data processing and analysis. The database will be checked by two members of the research team to avoid possible typing errors or inconsistencies in the programme.

Data will be expressed as means with respective SDs, medians and frequencies. For continuous variables, Student's t-test (normal distribution) or Mann-Whitney-Wilcoxon test (non-normal distribution) will be used in comparing the experimental and the sham groups; and for categorical variables, the X^2 or Fisher's exact tests will be employed. For dichotomous variables, McNemar's test or Bowker's symmetry test for categorical variables will be used in comparing parameters within groups (before vs after intervention). Analysis of variance (ANOVA) for repeated measures will be used to obtain between groups, within groups and interaction effects (groups vs times) for continuous variables, for example, the NRS pain level. For intergroup comparisons, ANOVA will be used followed by Tukey's test, and by contrast profile test for intragroup comparisons. The analysis will be performed by intention to treat.

Monitoring

Trial data integrity will be monitored by regularly scrutinising data files for omissions and errors. There are almost no adverse effects with the specified parameters.[2] The PBM in the red-light spectrum may cause discomfort due to an increase in local temperature; if it is intense, the intervention will be interrupted. The tolerated dose will be registered.

There will be one coordinating investigator for each centre (TAAdM or SLN) and a senior investigator not involved in the day-to-day administration of the trial (FGS), to ensure quality of data collection and monitoring with weekly auditing of data collection.

This project was evaluated by both hospitals' safety committees, which is routine in both services in this study.

Patient and public involvement

None.

ETHICS AND DISSEMINATION

This research protocol was approved by the Research Ethics Committees of the University of Campinas, Brazil (number CAAE: 59400922.1.1001.5404) and of the School Maternity Assis Chateaubriand, Brazil (number CAAE: 59400922.1.3001.5050). All participants can participate only after signing the informed consent form for PT arm (online supplemental material) or the informed consent form for NT arm (online supplemental material). The consent form contains the purpose of this study, procedures that will be performed, and potential benefits and risks of the intervention. Protocol amendments will be numbered and uploaded to the trial site. Participants will remain anonymous and will be given an affirmation of confidentiality and protection of the data collection. All authors declare there are no conflicts of interest, will be guaranteed the right to compensation in the event of any damage demonstrably resulting from the research, under the responsibility of the researchers. We will publish the results in a peer-reviewed journal.

DISCUSSION

Considering the possible benefits of the single application of PBM in pain relief and wound healing, this resource could benefit the puerperal woman, facilitating mobility, self-care, newborn care activities and breast feeding. It is an easy-to-apply resource with probably and ideally few side effects.

The use of low-level laser has been used for several medical issues, with analgesic, anti-inflammatory and healing effects, but few studies evaluate its effectiveness for perineal and nipple pain, in addition to the cost-effectiveness of its application. Our objective is to test an effective dose of PBM to manage pain in the immediate postpartum period. Our study limits the evaluation to the immediate effects of PBM but does not investigate its long-term or successive applications on healing, pain intensity, sexual function and breastfeeding success.

We chose to test the effects of a single application of laser therapy in the first 36 hours after childbirth, as previous studies demonstrate that pain is present even during this early postpartum period.[9 10 23 27] Perineal pain may impact the mother's mobility, self-care and care of the newborn.[23] Perineal pain tends to reduce between 4

and 10 days after childbirth for most women, regardless of perineal trauma.[27]

The first hours after birth is a critical period for breast feeding, the breastfeeding woman is learning how to make the correct approach and to handle this new ability. Early NT is a barrier to breast feeding and breast engorgement might perpetuate this difficulty.[28]

Interventions to control pain (perineal or nipple) even during hospitalisation during childbirth are even more important in Brazil and in other places where women, for different reasons, have difficulty accessing health services in the postpartum period. Thus, the period immediately after birth, even during hospitalisation, is a window of opportunity to treat NT or PT in women who present these complications early. New studies may address later interventions in the future.

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